

Doom PWAD QA Report

Run e9d3769d-558f-47b7-ae2e-f123a21c4209 | Map E1M1 | Outcome player_died

Executive Summary

This is an autonomous QA test of map E1M1 using an LLM-directed test harness. The test was designed to evaluate map navigation, combat engagement, resource distribution, and secret accessibility, resulting in a player death outcome after 68 seconds of runtime.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL

overall_verdict: FAIL

- The run ended with outcome 'player_died' after visiting 11 coarse position cluster(s).
- 2 kill(s) recorded against 25 enemy/enemies that spawn at difficulty 3.
- Selected-difficulty health_ratio=0.1909, ammo_ratio=0.1848.
- 0 secret(s) found.

No technical problems encountered during this test run.

Test Item And Environment

The test focused on the E1M1 map within the Freedoom1 IWAD at difficulty level 3. The map contains 69 enemy instances, 78 items, 3 keys, and 8 secret sectors, spread across a 5952x3712 unit space. The static analysis indicates high map complexity with a focus on hitscanner combat (81.16% hitscanner density).

The test environment utilized the ViZDoom engine, the gemini-3.1-flash-lite model, and an MCP interface. The run was executed at difficulty level 3 with a maximum tick capacity of 5000.

Metric	Value	Metric	Value
Status	completed	Duration	68
Actions	12	LLM calls	12
Final HP	0	Kills	2
Position samples	13	Movement units	6601.1
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/e9d3769d-558f-47b7-ae2e-f123a21c4209.mp4		

Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	69	25	44 hidden by skill/multiplayer flags
Items	78	71	7 hidden by skill/multiplayer flags
Secrets	8	8	Sector special 9
Geometry	571 linedefs	99 sectors	5952 x 3712 units

Selected-difficulty estimate: hard. Health ratio: 0.1909. Ammo ratio: 0.1848.

Coverage Evaluation

The automated playthrough covered approximately 11 position clusters across the map, indicating moderate exploration. However, with only 2 kills against 25 spawned enemies and 0 secrets found, the coverage was insufficient to fully evaluate the map's combat balance or secret-gating mechanisms.

Objectives Covered

- **Engaged initial enemy encounters:**
- **Basic map exploration within 11 position clusters:**

Objectives Omitted

- **Navigate the entire map:** Playthrough terminated at tick 2205 due to death
- **Find secrets:** 0 secrets found despite 8 secret sectors present in static analysis
- **Stress-test geometry:** Playthrough did not reach the majority of map sectors or complex interactive elements

Uncovered attributes: The playthrough could not test map-specific scripting, advanced secret sector triggers, or late-map combat encounters due to the termination of the run at the early-to-mid stage of the map.

Defects And Risks

Three defects were detected: one critical, one major, and one minor. The most impactful issue was a navigation/resource accessibility failure resulting in a critical-severity death.

Patterns: Defects suggest a correlation between navigation constraints and resource access, as the automated playthrough did not path to health items during critical moments.

Severity	Priority	Type	Tick	Description
2	2	agent_observed_resource_balance	2204	Automated playthrough observed resource balance issue [RESOURCE BALANCE] At tick 2204, position (2428.0, -3056.2): Health is 6 and the player has been unable to reach the visible Stimpack (ID 1) for several hundred ticks, likely due to geometry or navigation constraints. Player impact: imminent death. Severity: critical.
3	2	difficulty_spawn_mismatch	0	Difficulty-specific thing flags hide map content At difficulty 3, 44 of 69 enemies and 7 of 78 items from raw static analysis do not spawn in single-player because of Doom skill or multiplayer flags.
3	3	unreachable_secret	2205	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Position (2428.0, -3056.2):** high Critical health item appears unreachable or pathfinding fails to resolve interaction, leading to player death.

Activity Log Summary

1) Static analysis phase (tick 0), 2) Exploration and initial combat (ticks 18-1017), 3) Resource collection attempts (ticks 1595-2204), 4) Death and termination (tick 2205), 5) Report generation phase.

Exploration was interrupted significantly by the need for resource recovery, which ultimately failed to secure the automated playthrough's survival.

Tick	Event	Position	HP	MCP Action	Stop reason
203	kill	1922.9, -3606.6	100	set_objective	objective_set
402	item_pickup	2237.5, -2653.7	95	set_objective	objective_set
616	damage_taken	2170.8, -2742.0	68	get_situation_report	situation_report
809	damage_taken	2195.6, -2866.6	38	set_strategy	strategy_updated
1017	kill	1077.2, -2531.9	38	set_objective	objective_set
2204	item_pickup	2428.0, -3056.2	6	set_objective	objective_set
2205	death	2323.1, -3216.0	0	get_situation_report	episode_finished

Limitations And Recommendations

No significant test limitations identified beyond the premature termination of the run.

Good Quality Areas

- **Starting sector:** Automated playthrough successfully initiated combat and performed initial exploration without immediate errors.